SIMATS ENGINEERING

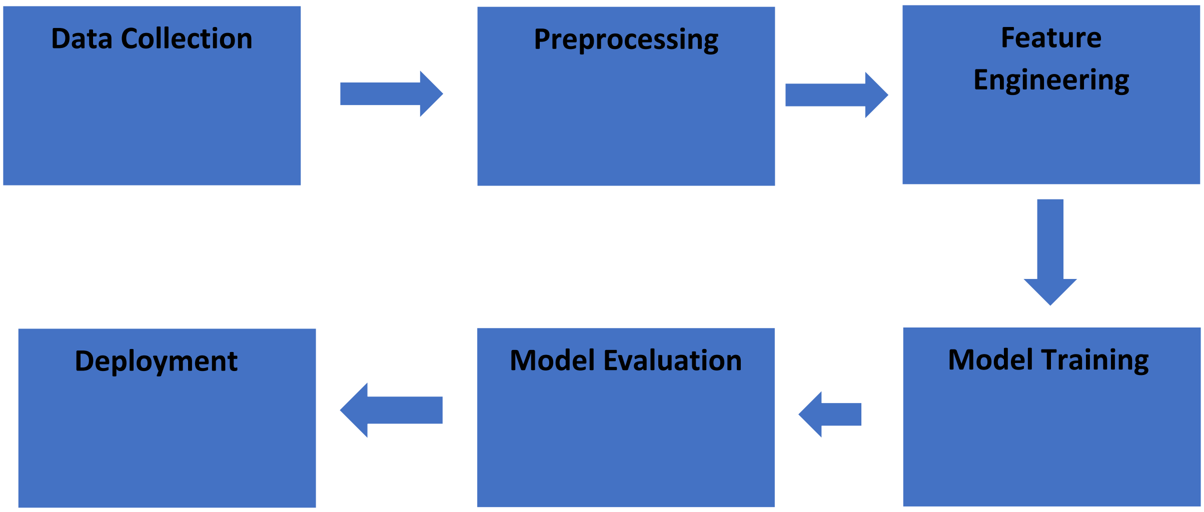
ASSESSMENT TOOL 2

COURSE CODE: ITA 0604 **COURSE NAME:** MACHINE LEARNING

CONCEPT MAPPING

1. **Explain the architecture of a typical machine learning pipeline with a neat diagram. Discuss the role of each stage including data collection, preprocessing, feature engineering, model training, evaluation, and deployment. Also, differentiate between training error and generalization error with suitable examples.**

Given:



1. Data Collection

Data is gathered from different sources such as databases, sensors, websites, or APIs. The quality and quantity of collected data directly affect model performance.

2. Data Preprocessing

Raw data is cleaned before training by:

- Removing missing values

- Removing duplicate records
- Handling noisy data
- Normalizing and scaling features

3. Feature Engineering

Important features are selected or created to improve prediction accuracy.

Examples include:

- Feature Selection
- Feature Extraction
- Encoding categorical variables

4. Model Training

The machine learning algorithm learns patterns from the training dataset.

Common algorithms include:

- Decision Tree
- Naïve Bayes
- Logistic Regression
- Support Vector Machine

5. Model Evaluation

The trained model is tested using unseen test data.

Common evaluation metrics:

- Accuracy
- Precision
- Recall
- F1-Score

6. Deployment

The trained model is deployed into real-world applications where it predicts outcomes for new data.

Examples:

- Spam Email Detection
- Face Recognition
- Product Recommendation

Example

A spam classifier achieves:

Training Accuracy = 99%
Testing Accuracy = 90%

Training Error

= 1%

Generalization Error

= 10%

A large difference indicates overfitting.

Final Conclusion

A good machine learning model should have low training error and low generalization error, ensuring good performance on unseen data.

2. Design a machine learning model for spam email classification using labeled data.
(a) Explain the choice of algorithm (e.g., Naïve Bayes, SVM, or Logistic Regression).
(b) Describe the preprocessing steps such as tokenization, stop-word removal, and vectorization.
(c) Discuss evaluation metrics such as accuracy, precision, recall, and F1-score.

Given:

(a) Choice of Algorithm

Algorithm: Naïve Bayes

Naïve Bayes is widely used for spam email classification because:

- Fast and efficient
- Works well with text data
- Handles high-dimensional features
- Requires less training data

Other possible algorithms include:

- Logistic Regression
- Support Vector Machine (SVM)

(b) Preprocessing Steps

Step 1: Tokenization

Split the email into individual words.

Example:

"Win a free prize today"

↓

Win | a | free | prize | today

Step 2: Stop-word Removal

Remove common words that do not contribute to classification.

Example:

Win free prize today

Step 3: Vectorization

Convert text into numerical form using:

- Bag of Words (BoW)
- TF-IDF

Example

Word	Frequency
Win	1
Free	1
Prize	1
Today	1

(c) Evaluation Metrics

Accuracy

Measures overall prediction correctness.

Precision

Measures how many predicted spam emails are actually spam.

Recall

Measures how many actual spam emails are detected.

F1-Score

Balances Precision and Recall.

Conclusion

Naïve Bayes combined with proper preprocessing provides high accuracy and fast spam detection.

3. In the context of the Candidate-Elimination Algorithm, consider a dataset with attributes:

Sky, AirTemp, Humidity, Wind, Water, Forecast and target concept EnjoySport.

(a) Explain the initialization of S (specific boundary) and G (general boundary).

(b) Show how S and G change with each training example.

(c) Compare Candidate-Elimination with the Find-S algorithm in terms of generalization capability.Specificity of the classifier, explaining its effectiveness in medical diagnosis.

(a) Initialization of S and G

Specific Boundary (S)

Initially,

S = <Ø, Ø, Ø, Ø, Ø, Ø>

Represents the most specific hypothesis.

General Boundary (G)

Initially,

G = <?, ?, ?, ?, ?, ?>

Represents the most general hypothesis.

(b) Updating S and G

Training Example 1 (Positive)

Sunny, Warm, Normal, Strong, Warm, Same → Yes

Update:

S = <Sunny, Warm, Normal, Strong, Warm, Same>

G = <?, ?, ?, ?, ?, ?>

Training Example 2 (Positive)

Sunny, Warm, High, Strong, Warm, Same → Yes

Generalize differing attribute:

S = <Sunny, Warm, ?, Strong, Warm, Same>

Training Example 3 (Negative)

Rainy, Cold, High, Strong, Warm, Change → No

Specialize G:

G =
<Sunny, ?, ?, ?, ?, ?>
<?, Warm, ?, ?, ?, ?>

S remains unchanged.

(c) Candidate-Elimination vs Find-S

Candidate-Elimination	**Find-S**
Maintains S and G boundaries	Maintains only S
Learns all consistent hypotheses	Learns one most specific

	hypothesis
Handles positive and negative examples	Uses only positive examples
Better generalization	Limited generalization
More accurate but computationally expensive	Simpler and faster

Conclusion

Candidate-Elimination is more powerful than Find-S because it considers both positive and negative examples and maintains the complete version space, resulting in better generalization and improved learning accuracy.